

TMR 4190 FINITE ELEMENT METHODS IN STRUCTURAL ANALYSIS

AUTUMN 2026

EXERCISE 6

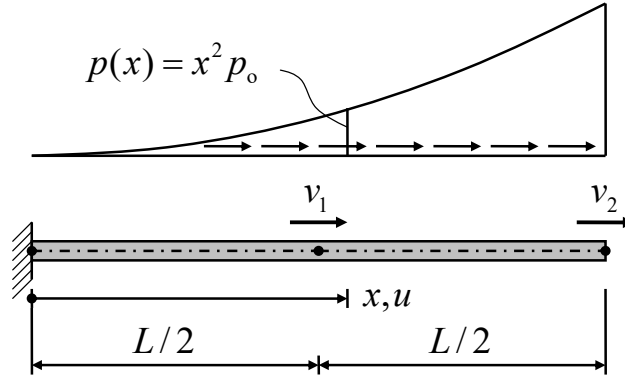


Figure 1

Figure 1 shows a bar element with three nodes. The bar is fixed at the left node ($x = 0$), while the other two nodes are free. The bar has a section area A , length L and elasticity modulus E . The bar is subjected to an external load in the longitudinal direction $p(x) = x^2 p_0$.

- a) Assume that the displacement $u = u(x)$ is expressed in terms of generalized coordinates

$$u = u(x) = xq_1 + x^2q_2 = [x \quad x^2] \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \mathbf{N}_q \mathbf{q}$$

$$\varepsilon_x = \frac{du}{dx} = \mathbf{B}_q \mathbf{q}$$

Compute $\mathbf{B}_q$, $\mathbf{k}_q = EA \int_0^L \mathbf{B}_q^T \mathbf{B}_q dx$ and $\mathbf{S}_q^o = - \int_0^L \mathbf{N}_q^T p(x) dx$, and solve for $\mathbf{q}$.

Determine the solution for the displacement $u = u(x)$ and the stress $\sigma_x = \sigma_x(x)$.

- b) Now, the displacement u is expressed with the Lagrange polynomials

$$u = N_1 v_1 + N_2 v_2 = [N_1 \quad N_2] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \mathbf{N} \mathbf{v}$$

where N_1 and N_2 are the second and third Lagrange polynomials of order 2 (the first polynomial is suppressed with the relative dof of the left node, because of the boundary conditions).

$$\varepsilon_x = \frac{du}{dx} = \mathbf{B} \mathbf{v}$$

Compute $\mathbf{N}$, $\mathbf{B}$, $\mathbf{k} = EA \int_0^L \mathbf{B}^T \mathbf{B} dx$, and $\mathbf{S}^o = - \int_0^L \mathbf{N}^T p(x) dx$, and solve for $\mathbf{v}$.

Determine the solution for the displacement $u = u(x)$ and the stress $\sigma_x = \sigma_x(x)$. Check if it is the same as in a).

Hint: you can use either x or ξ as coordinates (ξ is defined over the length of the bar with $\xi \in [-1, 1]$).

- c) Establish the **A**-matrix that defines the relationship between the generalized coordinates **q** and the nodal displacements **v**, i.e., $\mathbf{v} = \mathbf{A}\mathbf{q}$, and show that $\mathbf{k} = \mathbf{A}^{-T}\mathbf{k}_q\mathbf{A}^{-1}$.
- d) Derive the exact analytical expressions for the displacement $u(x)$ and the axial stress $\sigma(x)$ under the distributed load $p(x) = x^2 p_o$. Plot the exact and finite element solutions and explain why they differ. For a constant distributed load $p(x) = p_o$, would the finite element solution be exact? Justify your answer.